THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2241 Introduction to Mathematical Analysis
Test 1 (Suggested Solution)

1. Firstly, for any $m, n \in \mathbb{N}$, we have

$$\frac{m}{m+n} < \frac{m+n}{m+n} = 1.$$

Therefore, 1 is an upper bound of A. Since A is a nonempty subset of $\mathbb{R}$ which is bounded above, by the Completeness Axiom, the supremum exists. We claim that $\sup A = 1$.

Since 1 is an upper bound of A, we will prove that for any $\epsilon > 0$, there exists $a \in A$ such that $a > 1 - \epsilon$. Then by *Theorem 1.27*, we have $\sup A = 1$.

For any $\epsilon > 0$, by the Archimedean Property, there exists $N \in \mathbb{N}$ such that $\frac{1}{N+1} < \epsilon$. Then we have $1 - \frac{1}{N+1} > 1 - \epsilon$ and hence

$$\frac{N}{N+1} > 1 - \epsilon.$$

Now note that $\frac{N}{N+1}$ is in the set A with $m = N$ and $n = 1$. Thus we can take $a = \frac{N}{N+1}$ which satisfies $a > 1 - \epsilon$. The proof is complete.

2. (This question is very similar to *Assignment 2, Q.20(a)* and can be regarded as a generalization of it.)

   (a) For any $n \in \mathbb{N}$,

$$x_{n+2} - x_{n+1} = (1-c)x_{n+1} + cx_n - x_{n+1} = c(x_n - x_{n+1}) = -c(x_{n+1} - x_n).$$

   This implies $|x_{n+2} - x_{n+1}| = c|x_{n+1} - x_n|$. It follows that for any $n \in \mathbb{N}$,

$$|x_{n+2} - x_{n+1}| = c|x_{n+1} - x_n| = c^2|x_n - x_{n-1}| = \cdots = c^n|x_2 - x_1| = c^n|\beta - \alpha|.$$

   (b) Note that if $\alpha = \beta$, then (a) implies that $\{x_n\}$ is a constant sequence and hence must be a Cauchy sequence. Thus we may assume that $\alpha \neq \beta$. For any $n > m \geq 1$,

$$\begin{aligned}
|x_n - x_m| &= |(x_n - x_{n-1}) + (x_{n-1} - x_{n-2}) + \cdots + (x_{m+1} - x_m)| \\
&\leq |x_n - x_{n-1}| + |x_{n-1} - x_{n-2}| + \cdots + |x_{m+1} - x_m| \\
&= (c^{n-2} + c^{n-3} + \cdots + c^{m-1})\,|\beta - \alpha| \qquad \text{(by (a))} \\
&= c^{m-1}(c^{n-m-1} + c^{n-m-2} + \cdots + 1)\,|\beta - \alpha| \\
&= c^{m-1}\left(\frac{1 - c^{n-m}}{1-c}\right)|\beta - \alpha| \\
&< c^{m-1}\left(\frac{1}{1-c}\right)|\beta - \alpha| = Kc^{m-1},
\end{aligned}$$

   where $K = \frac{|\beta - \alpha|}{1-c}$ is a positive constant.

Now since $0 < c < 1$, the sequence $\{c^m\}$ converges to 0. Thus for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for any $m \in \mathbb{N}$, if $m \geq N$,

$$c^m < \frac{c}{K}\,\epsilon, \quad \text{or equivalently,} \quad c^{m-1} < \frac{1}{K}\,\epsilon.$$

It follows that for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for any $n, m \in \mathbb{N}$, if $n > m \geq N$, then $|x_n - x_m| < \epsilon$.

(c) Since $\{x_n\}$ is a Cauchy sequence, it must converge to some limit L. Observe that for any integer $n \in \mathbb{N}$,

$$x_{n+2} + cx_{n+1} \;=\; (1-c)x_{n+1} + cx_n + cx_{n+1} \;=\; x_{n+1} + cx_n,$$

so we have

$$x_{n+2} + cx_{n+1} \;=\; x_{n+1} + cx_n \;=\; \cdots \;=\; x_2 + cx_1 \;=\; \beta + c\alpha.$$

Since

$$\lim_{n\to\infty} (x_{n+2} + cx_{n+1}) \;=\; \lim_{n\to\infty} x_{n+2} + c \lim_{n\to\infty} x_{n+1} \;=\; L + cL,$$

it follows that

$$L + cL = \beta + c\alpha, \text{ and hence } \; L = \frac{\beta + c\alpha}{1+c}.$$

3. (a) It is given that $0 < b_1 \leq a_1$. For any $k \geq 1$, assume that $0 < b_k \leq a_k$ is true. Now for $n = k+1$, by inductive hypothesis, we have $b_{k+1} = \frac{2a_kb_k}{a_k+b_k} > 0$ and

$$a_{k+1} - b_{k+1} \;=\; \frac{a_k + b_k}{2} - \frac{2a_kb_k}{a_k+b_k} \;=\; \frac{(a_k-b_k)^2}{2(a_k+b_k)} \;\geq\; 0.$$

Combining the results, we have $0 < b_{k+1} \leq a_{k+1}$.

(b) For any $n \in \mathbb{N}$,

$$a_{n+1} - a_n \;=\; \frac{a_n + b_n}{2} - a_n \;=\; \frac{b_n - a_n}{2} \;\leq\; 0 \quad \text{(by (a))}.$$

This shows that $\{a_n\}$ is decreasing. On the other hand, since all terms are positive, the quotient $\frac{b_{n+1}}{b_n}$ is well-defined and

$$\frac{b_{n+1}}{b_n} \;=\; \frac{2a_nb_n}{(a_n+b_n)b_n} \;=\; \frac{2a_n}{a_n+b_n} \;=\; \frac{a_n}{\frac{a_n+b_n}{2}} \;=\; \frac{a_n}{a_{n+1}} \;\geq\; 1.$$

This shows that $\{b_n\}$ is increasing.

By (a), we see that $\{a_n\}$ is bounded below by 0. Furthermore, for any $n \in \mathbb{N}$, we have $b_n \leq a_n \leq a_1$ as $\{a_n\}$ is decreasing. Hence $\{b_n\}$ is bounded above by a_1. Now by the Monotone Convergence Thoeorem, we can conclude that both $\{a_n\}$ and $\{b_n\}$ converge.

(c) Suppose that $\{a_n\}$ converges to L_1 and $\{b_n\}$ converges to L_2. By the definition of a_{n+1}, we have

$$L_1 \;=\; \lim_{n\to\infty} a_{n+1} \;=\; \lim_{n\to\infty} \frac{a_n + b_n}{2} \;=\; \frac{L_1 + L_2}{2},$$

which implies $L_1 = L_2$. Denote by L this common limit. Now for any $n \in \mathbb{N}$, consider the term $a_{n+1}b_{n+1}$:

$$a_{n+1}b_{n+1} \;=\; \frac{a_n + b_n}{2} \cdot \frac{2a_nb_n}{a_n+b_n} \;=\; a_nb_n.$$

From this we obtain

$$a_{n+1}b_{n+1} \;=\; a_n b_n \;=\; \cdots \;=\; a_1 b_1.$$

Then

$$L^2 \;=\; \lim_{n\to\infty} a_{n+1} \lim_{n\to\infty} b_{n+1} \;=\; \lim_{n\to\infty} a_{n+1}b_{n+1} \;=\; \lim_{n\to\infty} a_1 b_1 = a_1 b_1.$$

Since all terms are positive, it follows that $L = \sqrt{a_1 b_1}$.

4. (a) True

First of all, $\sup A$ exists by the Completeness Axiom. Next, B is nonempty as A is bounded above. Moreover, B is bounded below by any element of A. Hence, by the Completeness Axiom again, $\inf B$ exists. It remains to show $\inf B = \sup A$.

For any $b \in B$, we have $\sup A \le b$ since $\sup A$ is the least upper bound for A. Therefore, $\sup A$ is a lower bound for B. Moreover, $\sup A \in B$ because $\sup A$ is an upper bound for A. This implies $u \le \sup A$ for any lower bound u of B. It follows that $\sup A$ is the greatest lower bound for B and hence the result.

(b) False

Define $a_n = 0$ if n is odd, and $a_n = n$ if n is even. So we have

$$\{a_n\} = \{0, 2, 0, 4, 0, 6, 0, 8, 0, \dots\}.$$

$\{a_n\}$ is NOT bounded above: By the Archimedean Property, for any $M \in \mathbb{R}$, there exists $N \in \mathbb{N}$ such that $2N > M$. Thus for any $M \in \mathbb{R}$, we have $a_{2N} = 2N > M$.

$\{a_n\}$ has a convergent subsequence: The subsequence of odd terms is just the constant sequence $\{0, 0, 0, 0, \dots\}$, which is clearly convergent.

(c) True (**Bookwork** – see *Assignment 1, Q.16*)

First of all, the set $\{a_n : n \in \mathbb{N}\}$ is nonempty and bounded above, so its supremum exists. Similarly, the infimum of $\{b_n : n \in \mathbb{N}\}$ exists. Write $a = \sup\{a_n : n \in \mathbb{N}\}$ and $b = \inf\{b_n : n \in \mathbb{N}\}$.

" $\subseteq$ ": Suppose $x \in I_n$ for all $n \in \mathbb{N}$. Then we have $a_n \le x \le b_n$ for all $n \in \mathbb{N}$. It follows that x is an upper bound for the set $\{a_n : n \in \mathbb{N}\}$ and so $a \le x$. Similarly, we have $x \le b$. Hence $x \in [a, b]$.

" $\supseteq$ ": Suppose $x \in [a, b]$. Since $a_n \le a$ and $b \le b_n$ for all $n \in \mathbb{N}$, we have $a_n \le a \le x \le b \le b_n$, that is, $x \in I_n$ for all $n \in \mathbb{N}$.